AERO70024 Applications of Computational Fluid Dynamics

Course Assignment 2025-26

1 Overview

This coursework objective is to produce high-fidelity simulation of incompressible flow over the *Avistar* airfoil. The goal is to obtain the aerodynamic coefficients, represent the result in the form of a lift-drag polar and compare the results to the data from experiments, XFOIL and STAR-CCM+ data. To do so, you will need to perform several 2D and a few Quasi-3D simulations following the instructions below to obtain and analyse the results.

1.1 Summary of Tasks

Theory

- Describe the differences between the Continuous Galerkin (CG) and Discontinuous Galerkin (DG) methods within the spectral/*hp* element framework. Comment on their function spaces, treatment of inter-element boundaries, stability properties, and computational cost. Discuss in which situations one method may be preferred over the other. This part should not exceed **2 pages**.

2D simulations

- Perform 2D simulations for angle of attacks $\alpha \in \{-5°,\ -3°,\ 3°,\ 6°,\ 10°\}$ with polynomial order $\mathscr{P} = (4, 5)$ for the pressure and velocity fields, respectively, at a Reynolds number of 402 100.
- For $\alpha = 3°$ perform a *h*-type and a *p*-type convergence study with at least 4 different data points for each case, to properly capture the convergence trend. Make sure you clearly define the error norm you are using and the reference case you consider. Plot both curves on the same figure. Discuss your findings.

Quasi-3D simulations

- Perform quasi-3D simulations for angle of attacks $\alpha \in \{-3°,\ 3°,\ 6°\}$ with $\mathscr{P} = (4, 5)$ at a Reynolds number of 402 100.
- Compare the vorticity field at a time instant after the flow has reached a statistically steady state for both 2D and Quasi-3D simulations and for $\alpha \in \{-3°,\ 3°,\ 6°\}$. Describe the differences you observe in the flow structures between 2D and Quasi-3D simulations.
- Compute C_L and C_D from both 2D and Quasi-3D simulations for all angles of attack simulated. Plot C_L vs α, C_D vs α and C_L vs C_D (C_D plot in the x-axis) and compare the results of 2D and 3D simulation with the experimental data, STAR-CCM and XFOIL in a single plot. Analyse your results.

An initial mesh and **incomplete** sessions files for both 2D and 3D simulations have already been prepared for you. The mesh works for both 2D and Quasi-3D simulations. These files can be accessed from the Blackboard.

1.2 General remarks and due date

- The due date for submitting your report in on **May 15th, 2026 at 23:59 UK time**.
- Your report should be at most **10 pages** (including everything, except for references) on A4 paper with 2*cm* margin using 11pt font.
- Figures should be neat, numbered and referred to in the report. They should have informative captions. Please avoid dark/black backgrounds, and ensure any annotations in the figures are easily readable.
- Do not repeat the instructions provided here for the simulation in your report. However, do include any additional filters or settings that you have learned from the user-guide/Jupyter notebook tutorials and used in your simulations.
- *It is fine to work in groups of 2 or a maximum of 3 to generate the data. However, all data post-processing and reports must be undertaken individually and you should declare who you worked with to produce the data.*
- Plan your work ahead. Note that the 2D simulations each will take about about 2-3 hours using 4 cores. On the other hand, the Quasi-3D simulations will take about 20 hours with 4 cores. **In order to avoid a saturation of the queuing system, please do not use more than 4 cores for your simulations.**
- Note that the computational node(s) have a limited number of cores. You must use the batch mode submission through the queuing system so that the nodes are used efficiently and in a fair manner for everyone.
- Read the following instructions carefully and consider the tutorial for the incompressible NS before starting your coursework.

2 Problem definition

The problem of interest is the simulation of turbulent flow at Re $= 402\,100$ over *Avistar* airfoil with chord length $c = 1$ at several angles of attack (α). The computational domain consists of a rectangular domain, where the airfoil leading edge is located at the $\mathbf{x} = (0, 0, 0)$ and extended in the x-direction with the length of $c = 1$ such that the airfoil chord length aligns with x-axis (See Fig. 1). The inflow and outflow boundaries are located $5c$ and $15c$ upstream and downstream with respect to the leading edge, respectively. The domain in the y-direction spans $10c$ on both sides with respect to the x-axis. The free stream flows into the domain at the inflow boundary with the velocity magnitude of U_∞ and with an angle of attack α° with respect to the airfoil chord, i.e. the x-axis as shown in Fig. 1.

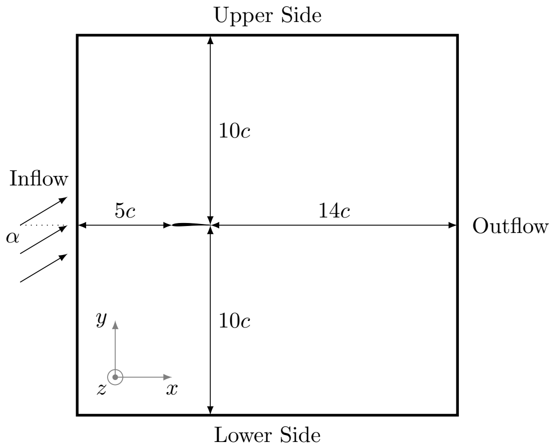
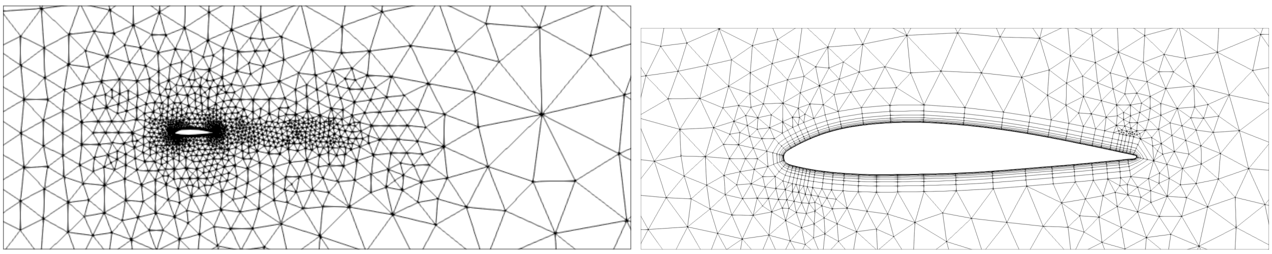


Figure 1: Schematic of the computational domain with the definition of the inflow condition and location of boundaries.

The fluid flow is governed by the non-dimensional incompressible Navier-Stokes equations. To conduct the simulations, the computational domain is tessellated with triangular and quadrilateral high-order elements. Figure 2a and 2b show part of the meshed domain. An initial mesh file has already been prepared and can be downloaded from BlackBoard. Recall that you can explore the mesh using Paraview.



(a) (b)

Figure 2: Two views of the mesh near the airfoil.

3 2D Simulations

A set of two–dimensional incompressible simulations is performed at a fixed Reynolds number Re $= 402\,100$, while varying the angle of attack according to the task specification. All simulations are defined through Nektar++ session files, which specify the mesh, polynomial expansions, boundary conditions, solver parameters, and post–processing filters. An incomplete session file is provided and completed following the guidelines below. Detailed instructions on session file setup can be found in the Nektar++ user guide.

3.1 Mesh composites and boundary definitions

The computational mesh contains several composites that group mesh entities into physical boundaries and fluid subdomains. Two composites define the fluid domain: one corresponding to triangular elements (ID $= 100$) and one to quadrilateral elements in the boundary layer (ID $= 101$). Additional composites define the airfoil surfaces (ID $= 5, 6$), inflow (ID $= 4$), outflow (ID $= 2$), and far–field boundaries (ID $= 1, 3$).

Boundary regions are assigned by mapping these composites to boundary IDs using the `BOUNDARYREGIONS` tag. Once defined, only boundary IDs are used when prescribing boundary conditions in the session file.

3.2 Non-dimensional formulation

The simulations are performed in non-dimensional form with chord length $c = 1$, free–stream velocity $U_\infty = 1$, and density $\rho = 1$. The kinematic viscosity is therefore

$$\nu = \frac{1}{402\,100},$$

and is specified in the session file using the `Kinvis` parameter.

3.3 Boundary conditions

A no–slip Dirichlet condition is imposed on the airfoil surface for the velocity, while a high–order Neumann condition is used for the pressure. Inflow and far–field boundaries are prescribed using the free–stream velocity corresponding to the chosen angle of attack. High–order outflow boundary conditions are applied at the outlet to avoid numerical instabilities.

3.4 Forces, time integration, and initialisation

The simulation is advanced using a fixed timestep $\Delta t = 10^{-4}$ and run until $T = 20$ to ensure steady time-averaged results. For efficiency recall to adapt the timestep based on the polynomial order used, accounting always for a CFL condition no larger than 0.3.

Initial conditions can be specified as the same as the inflow conditions with $p = 0$. If divergence is observed during the initial stages of the simulation, perform a low–order simulation first and use its final state to initialise the high–order run.

3.5 Mesh refinement file (`.mcf`)

For h–type refinement studies, the polynomial order is kept fixed and the mesh resolution is modified by adjusting these parameters in the `.mcf` file. More information on mesh generation can be found in this tutorial. Below is a brief overview of the main sections of the `.mcf` file, which is used to generate the 2D mesh. The main sections are:

- `INFORMATION`: defines the CAD geometry and mesh type.
- `PARAMETERS`: specifies global mesh sizing, polynomial order, and boundary–layer properties such as thickness, number of layers, and growth rate.
  - `EPS`: parameter to control non-linearly the element sizing, namely as $\propto \sqrt{\varepsilon(2-\varepsilon)}$.
  - `MinDelta`: minimum element size in the mesh.
  - `MaxDelta`: maximum element size in the mesh.
  - `Order`: order of the high-order mesh.
  - `BndLayerThickness`: thickness of the boundary layer mesh.
  - `BndLayerLayers`: number of layers in the boundary layer mesh.
  - `BndLayerProgression`: progression ratio for boundary layer mesh growth.
- `REFINEMENT`: introduces local mesh refinement regions, here defined along lines (of certain radius `R` and size of the elements `D`) around the airfoil to better resolve wake structures. Smaller values of `D` lead to finer meshes. Figure 3 illustrates the refinement region defined in the `.mcf` file.

To run the mesh generation, use the command:

```
NekMesh -v mesh.mcf mesh.xml
```

3.6 Filters

Aerodynamic forces acting on the airfoil are computed using the Nektar++ `AeroForces` filter. The filter integrates pressure and viscous stresses over the airfoil surface and outputs the force components in the Cartesian directions.

The filter is applied to the airfoil boundary by specifying its boundary ID in the session file:

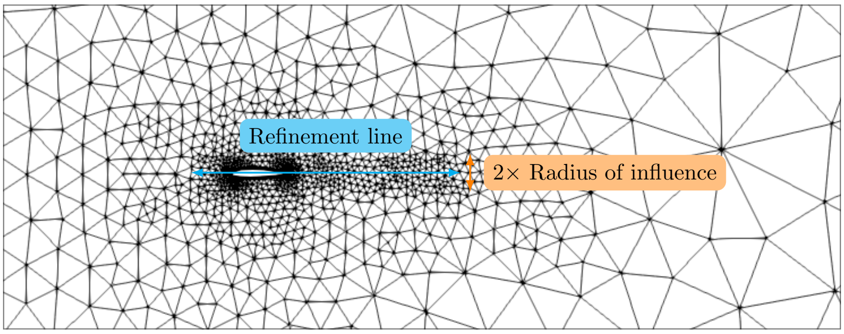


Figure 3: Schematic of the refinement region defined in the `.mcf` file.

```
<FILTER TYPE="AeroForces">
    <PARAM NAME="OutputFile">aeroForces</PARAM>
    <PARAM NAME="OutputFrequency">AeroForcesSteps</PARAM>
    <PARAM NAME="Boundary">B[boundary_number]</PARAM>
</FILTER>
```

The forces are written to the file `aeroForces.fce` at the specified output frequency. The filter outputs the force components F_x and F_y in the global coordinate system. Lift and drag forces are obtained by projecting these components onto the directions normal and parallel to the free–stream velocity, accounting for the angle of attack.

Since the simulations are performed in non-dimensional form with $\rho = 1$, $U_\infty = 1$, and $c = 1$, the resulting forces are already non-dimensional.

For h– and p–type convergence studies, it can be useful to monitor solution convergence at specific locations in the domain using history points. This can be achieved with the `HistoryPoints` filter, for example:

```
<FILTER TYPE="HistoryPoints">
    <PARAM NAME="OutputFile">HistoryPoints</PARAM>
    <PARAM NAME="OutputFrequency">AeroForcesSteps</PARAM>
    <PARAM NAME="plane">5,8,-2,-2,0,-2,2,0,5,2,0</PARAM>
</FILTER>
```

Further details on the use of history points are available in the Nektar++ documentation.

3.7 Running the 2D simulations using `slurm`

All 2D simulations are executed in batch mode using the `slurm` workload manager. A submission script is provided, which specifies the computational resources and runs the Nektar++ incompressible solver. The script assumes that the mesh and session files are named `mesh.xml` and `session.xml`, respectively, although these can be modified as needed inside the script.

To submit a simulation to the queue, the following command is executed from the directory containing the input files:

```
sbatch slurm.job
```

The output of the simulation is redirected to a log file, which can be monitored to check solver progress and stability. The status of submitted and running jobs can be inspected using:

```
squeue
```

If a simulation is submitted by mistake or needs to be terminated, it can be cancelled using:

```
scancel JOBID
```

where JOBID is the identifier reported by squeue.

For 2D simulations, the **short** queue is used, which is suitable for runs with wall times below six hours. Upon completion, the solver produces checkpoint files at regular intervals and a final solution file, which can be post-processed using FieldConvert for visualisation and analysis.

4 Quasi-3D Simulations

Quasi–3D (3DH1D) simulations are used to model spanwise flow development while assuming homogeneity in the z direction. In this approach, the flow is discretised in the z direction using Fourier modes, while the spatial discretisation in the x–y plane remains identical to the 2D case. The same 2D mesh is therefore reused for all Quasi–3D simulations.

4.1 Spanwise discretisation

The third dimension is introduced by defining the spanwise length L_Z and the number of Fourier planes HomModesZ in the session file. The number of Fourier planes must be even, with each pair forming one complex Fourier mode. For all Quasi–3D simulations, $L_Z = 0.25$ and HomModesZ $= 8$ are used.

To activate the Quasi–3D formulation, the solver is instructed to use a homogeneous direction by adding

```
<I PROPERTY="HOMOGENEOUS" VALUE="1D"/>
```

to the SOLVERINFO section of the session file.

4.2 Variables, expansions, and boundary conditions

In addition to the in–plane velocity components u and v, the spanwise velocity component w is introduced. The expansion settings are identical to the 2D case, with w included as an additional field.

No–slip boundary conditions are applied to all velocity components on the airfoil surface, while $w = 0$ is imposed on all remaining boundaries. Pressure boundary conditions are kept identical to the 2D configuration.

4.3 Initial conditions

Each Quasi–3D simulation is initialised using the final solution of the corresponding 2D simulation. The 2D velocity and pressure fields are loaded from the restart file, while the spanwise velocity component is initialised with a small perturbation to trigger three–dimensional instabilities:

```
<FUNCTION NAME="InitialConditions">
    <F VAR="u,v,p" FILE="mesh2d.fld" />
    <E VAR="w" VALUE="0.001*awgn(1.)" />
</FUNCTION>
```